Marawan Eldeib
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Software Engineering | IT Security

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Profile

Master's student in Information Technology at the University of Stuttgart, focusing on software engineering and IT security. Optimised NASA/DLR research software at the Institute of Space Systems and measured through HTTPS analysis which data LLM tools send to their servers. Seeking a full-time position in software engineering or IT security.

Education

since 10/2024 **M.Sc. Information Technology**

University of Stuttgart, focus on Software Engineering

- Master's thesis since 10/2026: A GitOps-Constrained Agentic Workflow for System Manipulation, in collaboration with the High Performance Computing Center Stuttgart, HLRS
- Relevant modules: Distributed Systems, Service Management and Cloud Computing, Deep Learning Lab

07/2020 – 10/2024 **B.Eng. (Hons.) Electronics Majoring in Computer**

Multimedia University, Cyberjaya, Malaysia

- CGPA 3.7/4.0, German grade approx. 1.4, First Class Honours
- Dean's List, Merit International Scholarship, 50 % of tuition fees

Work Experience

since 10/2026 **Student Research Assistant**

Institute for Control Engineering of Machine Tools and Manufacturing Units, ISW, University of Stuttgart

- Developing a software prototype in the INTERFACE4CPPS research project to integrate engineering artifacts for cyber-physical production systems

04/2026 – 08/2026 **Research Assistant, Software Development**

Institute of Space Systems, IRS, SOFIA Data Center, University of Stuttgart

- Got a NASA/DLR data-reduction job that failed on memory running again: peak 14.4 GB, output unchanged
- Root cause: redundant parallel computations; cut memory by 39 % via configuration and a further 32 % with memray

04/2025 – 12/2025 **Research Assistant**

Fraunhofer IOSB, Department of Information Management and Production Control, Karlsruhe

- Evaluated LLM-based data extraction for digital twins in Java and Spring Boot services
- Compared six large language models and built more than 30 evaluation datasets

06/2024 **AI Workshop Facilitator, contract**

Institute of Diplomacy and Foreign Relations, IDFR, Ministry of Foreign Affairs Malaysia

- Led AI and machine-learning workshops for more than 20 international participants
- Gave technical sessions on AI in agriculture, smart farming

07/2023 – 10/2023 **Software Engineering Intern, Mobile**

Ikhlas Com Sdn Bhd, AirAsia Group, Kuala Lumpur, Malaysia

- Fixed 14 defects and delivered 5 checkout-calculation features in a Flutter app with more than 10,000 downloads
- Refactored 2 modules and integrated GraphQL APIs for bilingual content management

Projects and Publication

04/2026 – 09/2026 **Measuring Data Exposure in LLM-Integrated Productivity Tools**

Python, mitmproxy | github.com/MarawanEldeib/privacy-analysis-research

- Measured outbound HTTPS and WebSocket traffic using planted canaries
- Two writing-assistant extensions sent all 12 planted secrets to their servers on paste

11/2024 – 02/2025 **Decentralized P2P Chat System, team project**

Python, Distributed Systems | github.com/ayushmittalde/chat_app_DS

- Built fault-tolerant peer-to-peer messaging in a six-person team
- Causal and total ordering with vector clocks, leader election with the Bully algorithm
- Reliable multicast and recovery from node crashes

11/2023 – 07/2024 **MangoVision, Object Detection in Aerial Images**

PyTorch, YOLOv8, Faster R-CNN | Bachelor research project

- Award: Best Project in Engineering, CS & IT, MMU Career Fair 2024
- Publication: “Performance Analysis of Faster R-CNN and YOLOv8 Model for Mango Fruits Detection”, JIWE, 2026, doi.org/10.33093/jiwe.2026.5.2.17

Skills

Programming	*very good:* C/C++, Dart, SQL, JSON; *good:* Python, TypeScript, HTML/CSS; *basic:* Java
Frameworks	*very good:* Flutter, REST; *good:* Spring Boot, GraphQL, React, Next.js, Node.js
Databases	*very good:* MySQL; *good:* PostgreSQL; *basic:* Redis
Linux & DevOps	*very good:* Ubuntu, Kali Linux, Pop!_OS, Bash, SSH, Slurm, Docker, Git, GitHub Actions, Playwright; *basic:* Kubernetes, Terraform, Ansible, AWS
IT Security	*very good:* mitmproxy, Wireshark, Burp Suite, OWASP ZAP; *good:* OWASP Top 10, Nmap, CTF
Engineering	*good:* Design Patterns, System Design, Distributed Systems, Algorithms and Data Structures
AI	*very good:* Claude Code, MCP; *good:* PyTorch
Methods	*very good:* Agile/Scrum, Jira, Confluence
Languages	Arabic: native, C2 | English: C1 | German: B1

Volunteering

08/2024 – 10/2024 **Volunteer tutor** for Digital Logic Design, Multimedia University

12/2022 – 12/2023 **Head of Membership & Student Affairs**, IEEE Student Branch MMU